MTH309

$F_1 \cap F_2$: sigma field, but $F_1 \cup F_2$ need not be sigma field : consider $F_1 = \sigma(\{A\})$, and $F_2 = \sigma(\{B\})$, as $\sigma(\mathscr{A})$: is the mnimal sigma field containing $\mathscr{A}$ which is an arbitrary collection of subsets of Ω.

$\mathscr{A}_2 \subseteq \sigma(\mathscr{A}_1)$ & $\mathscr{A}_1 \subseteq \sigma(\mathscr{A}_2) \Rightarrow \sigma(\mathscr{A}_1) = \sigma(\mathscr{A}_2)$

This is useful in defining $\mathscr{B}_{\mathbb{R}}$: sigma(collection of all open sets in $\mathbb{R}$) : Now as *open sets in* $\mathbb{R}$ are the finite and countable unions of open intervals in $\mathbb{R}$.

Open intervals in $\mathbb{R}$: $\{(a,b) : -\infty \leq a < b \leq \infty\}$. Let collection of all open intervals in R be G and collection of all open sets be H ⇒ $G \subseteq \sigma(G) \subseteq \sigma(H)$ and $H \subseteq \sigma(G)$ (by defn of H) ⇒ $\mathscr{B}_{\mathbb{R}} = \sigma(H) = \sigma(G)$.

Probability Measure

💡 Let $(\Omega, \mathscr{F})$ be a measurable space(i.e. $\mathscr{F}$ is a σ field. A measure is a non negative function $\mu : \mathscr{F} \to [0, \infty)$ that follows ***countable additivity.*** $(\Omega, \mathscr{F}, \mu)$ is a measure space.

Let μ be a measure defined on $(\Omega, \mathscr{F})$, with $\mu(\Omega) = 1$ ⇒ we say μ is a probability measure and $(\Omega, \mathscr{F}, \mu)$ is a probability space. An example is a ***dirac measure supported at point x.***

Dirac Measure on $\mathbb{R}$

Consider the non negative set function δ_x defined on $(\mathbb{R}, \mathscr{B}_R)$-

$\delta_x(A) = 1$ if $x \in A$, else 0. Note that $\delta_x(\mathbb{R}) = 1$.

Uniform measure on a discrete space

Let $\Omega = \{x_1, \ldots, x_n\}$ with a σ field 2^{Ω}. Let $\nu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}$ in the sense that for any $A \subset \Omega,\ \nu(A) = 1/n \sum_{i=1}^{n} \delta_{x_i}(A)$

💡 Note that any convex combination of probability measure is also a probability measure i,e, Let $\{\mu_n\}_n$ be countable measures and $\{\alpha_n\}_n$ be constants s.t. $\sum \alpha_i = 1$, then $\nu = \sum_{i=1}^{\infty} \alpha_i \mu_i$ is also a probability measure

Example define $\sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} \delta_k$ is a prob measure on $(\mathbb{R}, \mathscr{B}_R)$

Recall the properties of measure

- If $B \subseteq A$, then $\mu(A) = \mu(B) + \mu(A \cap B^c) \Rightarrow \mu(B) \leq \mu(A)$
- Inclusion Exclusion Formula
- ***Continuity Properties :*** Let $\{A_n\}_n$ be a sequence in $\mathscr{F}$ with either $A_n \uparrow A$ or $A_n \downarrow A$ then $lim_{n\to\infty}\mu(A_n) = \mu(A)$. In continuity from above it only holds when $\mu(A_k) < \infty$ for some k.
- the set $\mathscr{E} = \{A \in \mathscr{F} | \mathbb{P}_1(A) = \mathbb{P}_2(A)\}$ is a monotone class + is *closed under complementation and under disjoint unions(finite and countable)*

- a non negative set function with **Finite additivity + continuity** (from below or from above to ϕ) ⇒ Countable additivity = measure

Definition [σ finite measures] : If $\Omega_n \uparrow \Omega$, with $\Omega_n \in \mathscr{F},\ \forall n$, and $\mu(\Omega_n) < \infty,\ \forall n$ ⇒ μ is a σ finite measure.

> 💡 Note that the counting measure defined on $\mathbb{N}$ is infinite measure but it is sigma finite if $\Omega_n = \{1, 2, \ldots, n\}$

Borel Cantelli Lemma - First half

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. Let $\{A_n\}_n$ be a sequence in $\mathscr{F}$ with $\sum_n \mathbb{P}(A_n) < \infty$, then $\sum_n \mathbb{P}(limsup A_n)$ = 0

Measurable Functions

Let $(\Omega, \mathscr{F})$ be a measurable space and let f : $\Omega \to \mathbb{R}$. f is Borel Measurable if $f^{-1}(A) \in \mathscr{F},\ \forall A \in \mathbb{B}_R$.

Examples

- Indicator Function $1_B : \Omega \to \mathbb{R}$.
- Degenerate function $f(\omega) = c,\ \forall \omega \in \Omega$

> 💡 *Remark : Let* $f : \mathbb{R} \to \mathbb{R}$ be a function. Then f is continuous iff $f^{-1}(U)$ is open for all open subsets U of $\mathbb{R}$

> 💡 *Remark:* $f : \mathbb{R} \to \mathbb{R}$ *be a continuous function. then f is* ***Borel Measurable***

Random Variables

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. Any borel measurable function $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_R)$ is a random variable.

Definition[$\sigma(X)$] : $\sigma(X) = (X^{-1}(\mathbb{B}_R)) = \{X^{-1}(A) : A \in \mathbb{B}_R\}$, now as $X^{-1}(A) \in \mathscr{F},\ \forall A$ ⇒ $\sigma(X) \subseteq \mathscr{F}$ i.e. $\sigma(X)$ is a sub σ field of $\mathscr{F}$.

Law or Distribution of an RV

Consider $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}^d, \mathscr{B}_{R^d})$

$\mathbb{P} : \mathscr{F} \to [0, 1]$ and $\forall A \in \mathbb{B}_{R^d},\ X^{-1}(A) \in \mathscr{F}$

Thus, define $PoX^{-1} : B_{R^d} \to [0, 1]$

Examples :

- Conside the case of an degenerate RV : $PoX^{-1}(A) = 0,\ ifc \notin A$, else 1, thus essentially δ_c
- A bernoulli(p) RV has the law $p\delta_1 + (1-p)\delta_0$
- Binomail(n, p) has the law $\sum_{k=0}^{n} n_{C_k} p^k (1-p)^{n-k} \delta_k$

Thus, convex linear combinations of dirac measures = discrete RV

Caratheodery Extension Theorem

Given a measure μ on a field $\mathscr{F}$, we want to construct a measure ν on $\sigma(\mathscr{F})$ s.t. $\mu(A) = \nu(A),\ \forall A \in \mathscr{F}$

Theorem has two properties : Let μ be a σ finite measure on a field $\mathscr{F}$

Extension :

μ has an extension to a σ finite measure $\bar{\mu}$ on $\sigma(F)$ i.e. $\bar{\mu}(A) = \mu(A),\ \forall A \in \mathscr{F}$

Uniqueness :

If $\bar{\mu}$ and $\tilde{\mu}$ are 2 extensions as in the exsistence part then they are the same i.e. $\bar{\mu}(A) = \tilde{\mu}(A),\ \forall A \in \sigma(\mathscr{F})$ ← note!!

Read the 3 - stage proof in notes

Measure Theoretic Integration

Properties

1. **Say h is non-negative and integrable function,** then if $|g| <= h$ ⇒ g is integrable.
2. Let h be non-negative and measurable function then $\nu(A) = \int_A h d\mu$, is a measure for $A \in \mathscr{F}$.
   - If h is integrable then
   $\nu(\Omega) < \infty$ ⇒ *finite measure*
   -
   By **continuity of measure** (from below): if $A_m \uparrow A$, then $\int_{A_m} h d\mu \uparrow \int_A h d\mu$.
3. (*)

(*) - Monotone Convergence Theorem - MCT

Suppose $f_1, f_2, \ldots$ be **non-negative, measurable** functions such that $f_n \leq f_{n+1}$, and $f := lim_n f_n \geq 0$(also measurable), then $\int f_n d\mu \uparrow \int f d\mu$ i.e. $lim_n \int f_n d\mu = \int lim_n f_n d\mu$.

Extended Form MCT

Let $g, h_1, h_2, \ldots$ be Borel measurable functions s, then if either satisfies then

1. If $h_n \geq g, \forall n$ & $\int g d\mu > -\infty$ & $h_n \uparrow h$
2. If $h_n \leq g$ & $\int g d\mu < \infty$ & $h_n \downarrow h$

Then, $lim_n \int h_n d\mu = \int lim_n h_n d\mu$

💡 Note : $X^+,\ X^- < |X|$ (By property of measurable integrable function) , Now $\mathbb{E}|X| < \infty \iff \mathbb{E}X^+,\ \mathbb{E}X^- < \infty \iff \mathbb{E}X \in \mathbb{R}$

💡 ***Lebesgue vs Riemann Integral :*** Now let $f : \mathbb{R} \to \mathbb{R}$ be a function such that Riemann integral $\int_a^b f(x)dx$ exists, then the lebesgue integral i.e. $\int_{(a,b)} f d\lambda := \int_{\mathbb{R}} f 1_{(a,b)} d\lambda$ is equal to the riemann. Moreover note that riemann integral divides the area under the curve by vertical strips whereas the lebesgue integral is just the convergence from below of horizontal curves. Lebesgue integral gives a greater class of integrable functions.

μ - almost everywhere($\mathbb{P}$ - almost surely)

We say that a property $\mathscr{P}$ holds $\mu\ a.e.$($\mathbb{P}\ a.s.$) for a measurable function(or a collection of measurable functions) if $\mu(\omega : property \mathscr{P}$ does not hold for $\omega) = 0$

Some properties :

1. If f = 0 μa.e., then $\int f d\mu = 0$. Also if h ≥ 0 and $\int h d\mu = 0$, then $h = 0$ μ a.e.
2. $g = h$ μ a.e. ⇒ $\int g = \int h$, provided the integrals exist.

3. If h is μ integrable then h is finite μ a.e.

Fatou's Lemma

Let $f, f_1, f_2, \ldots$ be Borel- Measurable functions

1. If $f_n \geq f, \forall n$, and $\int f d\mu > -\infty$, then $\int \liminf_n f_n d\mu \leq \liminf_n \int f_n d\mu$
2. If $f_n \leq f$, and $\int f d\mu < \infty$, then $\int \limsup_n f_n \geq \limsup_n \int f_n d\mu$

DCT - Dominated Convergence Theorem

Let $f, f_1, f_2, \ldots$ be Borel Measurable functions s.t. $f_n \to f$ μ-a.e.. If $|f_n| <= g$, and g is integrable, $\forall n$, then f is **integrable and** $\int f d\mu = \int lim_n f_n d\mu = lim_n \int f_n d\mu$.

Further we can establish that $\lim_n \int |f_n - f| d\mu = 0$.

$\mathscr{L}^p$ - spaces

Definition

$\mathscr{L}^p(\mu) := \{f : \Omega \to \mathbb{R} | \text{f is measurable and } \int |f|^p d\mu < \infty\}$, define $||f||_p = (\int |f|^p d\mu)^{\frac{1}{p}}$.

DCT - Lp

Let $X, X_1, X_2, ..$ be random variables such that $X_n \to X$ a.s.. If $\exists$ RV $Y \in \mathscr{L}^p(\mathbb{P})$ s.t. $|X_n| \leq Y, \forall n$, then $X, X_1, \ldots \in \mathscr{L}^p(\mathbb{P})$, and $X_n \to X$ in Lp.

Properites :

1. Scalar Multiplication : If f belongs to Lp then αf belongs.
2. If $f = g$ μ a.e. $\Leftrightarrow |f - g| = 0$ μ a.e. $\Leftrightarrow \int |f - g|^p d\mu = 0 \Leftrightarrow ||f - g||_p = 0$